Reg. No. | | | | | | | | | | |

[12EE61/13EE61]

# SIXTH SEMESTER BE/B.TECH EXAMINATION, JUNE-2016

## POWER SYSTEM ANALYSIS – I

Time:3 Hours

Maximum Marks:70

**Instructions: Answer any FIVE full questions**

1. a. Define per unit quantity. Mention the advantages of per unit system. 04

   b. Explain the procedure of drawing per unit reactance diagram from single line diagram. 04

   c. Draw the reactance diagram of the system shown in figure 1c. The ratings of the components are shown in the figure. Choose generator rating as base values. 06

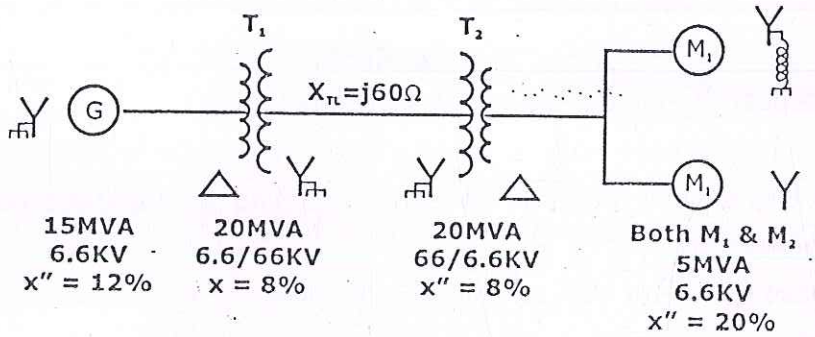



Fig. 1(c)

2. a. Explain in detail the transients on a transmission line. 06

   b. A synchronous generator and a motor are rated 30MVA, 13.2KV. both have sub transient reactance of 20%. The line connecting them has a reactance of 20%, on the base of machine rating. The motor is drawing 20MW at 0.8 pf (lead). The terminal voltage of motor is 12.8KV, when a symmetrical fault occurs at motor terminals, find the sub transient current in generator, motor and at the point of fault. 08

3. a. Show that power is invariant using symmetrical components. 07

   b. The positive, negative and zero sequence components of the currents are $20\angle 10°, 6\angle 60°$ and $3\angle 30°$ amp respectively. Determine the line currents. 07

4. a. Draw the positive, negative and zero sequence networks for the power system shown in fig 4(a). Choose a base of 50MVA, 220KV in the 50Ω transmission lines and mark all reactances in p.u. The ratings of the generators and transformers are, 08

   Generator 1: 25MVA, 11KV, x" = 20%

   Generator 2: 25MVA, 11KV, x" = 20%

Three phase transformer (each): 20MVA, 11Y/220Y KV, x=15%. The negative sequence reactance of each synchronous machine is equal to the Subtransient reactance. The zero sequence reactance of each machine is 8%. Assume that the zero sequence reactance of lines are 250% of their positive sequence reactances.

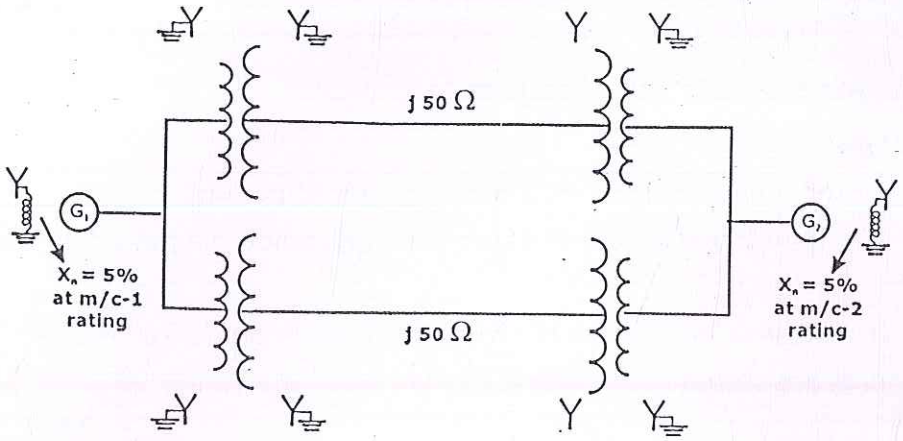


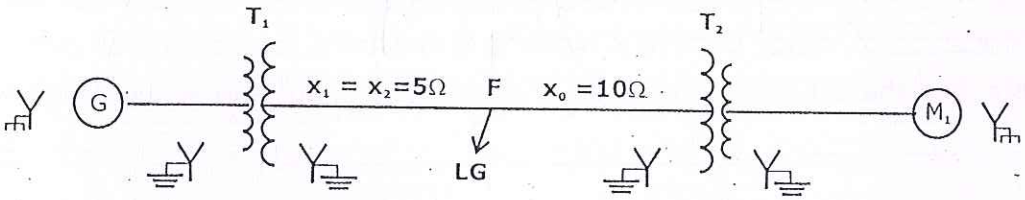

**Fig. 4(a)**

b. What are sequence impedances and sequence networks? **06**

5. a. Derive an expression for fault current, when line to line fault occurs on an unloaded generator. **07**

b. A three phase generator with an open circuit voltage of 400V is subjected to an SLG fault through a fault impedance of j2Ω. Determine the fault current, if $z_1$=j4Ω, $z_2$ = j2Ω and $z_0$ = j1Ω. Repeat trhe problem for LL fault. **07**

6. A synchronous motor is receiving 10MW of power at 0.8pf lag at 6KV. An LG fault takes places at the midpoint of the transmission line as in fig.6. Find the fault current the rating of the generator motor and transformer are as under. **14**




| | | | |
|---|---|---|---|
| 20MVA<br>11KV<br>$x_1$ = 0.2pu<br>$x_2$ = 0.1pu<br>$x_0$ = 0.1pu | 18MVA<br>11.5/34.5KV<br>X = 0.1pu | 15MVA<br>34.5/6.9KV<br>X = 0.1pu | 15MVA<br>6.9KV<br>x1 = 0.2pu<br>$x_2$ = $x_0$ = 0.1pu |

**Fig. 6**

7. a. Derive the swing equation for a single machine connected to an infinite bus. **07**

b. Differentiate between steady state stability and transient stability. **07**

8. a. Write short note on Equal Area Criterion. **07**

b. Explain any one application of equal area criterion. **07**